PROPERTY F IMPLIES WEAK INTEGRALITY FOR NON-DEGENERATE PIVOTAL BRAIDED FUSION CATEGORIES

Abstract. We prove that every non-degenerate pivotal braided fusion category with property F is pseudo-unitary and weakly integral. In particular, every modular category with property F is weakly integral. Neither sphericality nor pseudo-unitarity is assumed. Passing to the Drinfeld centre extends the result to degenerate categories whose dimensions are positive on the intersection of the Müger centre with the adjoint subcategory; these include every braided fusion category with a pseudo-unitary Galois conjugate.

The Frobenius–Perron dimension is the normalised S-matrix column of a simple object P, and unitarity of the normalised S-matrix makes the global dimension equal to $|d_P|^2$ times the Frobenius–Perron dimension. A trace inequality for finite braid group images shows that the double braiding of P with itself can attain its maximal size only if $|d_P| = \mathrm{FPdim}(P) \geq 1$, and this forces pseudo-unitarity.

1. Introduction

Let $\mathcal{C}$ be a braided fusion category over $\mathbb{C}$. For an object X and $n \geq 2$, write $G_{X,n}$ for the image of

$$\rho_{X,n} : B_n \longrightarrow \mathrm{Aut}(X^{\otimes n}), \qquad \sigma_i \longmapsto \mathrm{id}_X^{\otimes(i-1)} \otimes c_{X,X} \otimes \mathrm{id}_X^{\otimes(n-i-1)}.$$

The category has *property F* if every $G_{X,n}$ is finite, and is *weakly integral* if $\mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$.

Conjecture (Naidu–Rowell [NR])**.** *A braided fusion category has property F if and only if it is weakly integral.*

Thus the conjecture relates finiteness of all braid group images to a single arithmetic condition on the fusion category.

The implication from weak integrality to property F is known for weakly group-theoretical braided fusion categories [GN, Theorem 7.2], extending the group-theoretical case [ERW]. Etingof, Nikshych and Ostrik asked whether there exists a weakly integral fusion category that is not weakly group-theoretical [ENO2, Question 2]; a negative answer would establish that implication in general. In the other direction, infinite braid group images have been established for several families of quantum-group categories [FLW, LRW]. We prove the implication from property F

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to weak integrality for non-degenerate categories with a pivotal structure. Recall that $\mathcal{C}$ is *pseudo-unitary* if $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C})$.

Theorem 1.1. *Let $\mathcal{C}$ be a non-degenerate pivotal braided fusion category. If $\mathcal{C}$ has property F, then $\mathcal{C}$ and all its Galois conjugates are pseudo-unitary, and* $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$. *In particular, every modular category with property F is weakly integral.*

Equivalently, a non-degenerate pivotal braided fusion category that is not weakly integral admits a braid group representation with infinite image.

The Frobenius–Perron dimension is a character of the fusion ring. Since $\mathcal{C}$ is non-degenerate, it is the normalised S-matrix column of a simple object P, and unitarity of the normalised S-matrix gives $\dim(\mathcal{C}) = |d_P|^2 \,\mathrm{FPdim}(\mathcal{C})$. As $\dim(\mathcal{C}) \leq \mathrm{FPdim}(\mathcal{C})$, it suffices to show that $|d_P| \geq 1$. The double braiding of P with itself has the largest trace that a character allows. We show that finite braid images then force $|d_P| = \mathrm{FPdim}(P)$, which is at least one.

The main step is the trace argument in Section 3. It first bounds the categorical trace of every unitary in a braid algebra. The braid $\sigma_1^2\sigma_2^2$ on three strands attains this bound; the equality case makes $\sigma_1^2\sigma_2^{-2}$ act trivially wherever the trace sees it, and computing its trace in two ways gives the required identity of dimensions. Pivotal dimensions may be non-real, and the argument keeps track of their phases.

Section 2 records the transfer properties of property F and the facts about pivotal dimensions used in the proof, and constructs the Frobenius–Perron object. Section 3 establishes the trace bound and the saturation lemma. The proof of Theorem 1.1 is completed in Section 4. Section 5 passes to the Drinfeld centre. It extends the theorem to degenerate categories whose dimensions are positive on the intersection of the Müger centre with the adjoint subcategory, which includes every braided fusion category with a pseudo-unitary Galois conjugate. It then derives a criterion for a single object to have infinite braid images.

2. Preliminaries

In the sequel, all fusion categories are over $\mathbb{C}$, and we suppress associativity and unit constraints. We write $\mathbf{1}$ for the tensor unit, $\mathcal{O}(\mathcal{C})$ for the set of simple objects up to isomorphism, and $N_Z^W = \dim \mathrm{Hom}(W, Z)$ for the multiplicity of a simple object W in an object Z. For a fusion subcategory $\mathcal{A}$ of a braided category, $\mathcal{A}'$ denotes its centraliser; thus $\mathcal{C}'$ is the Müger centre, and $\mathcal{C}$ is non-degenerate if $\mathcal{C}' = \mathrm{Vec}$. We write $\mathcal{C}_{\mathrm{ad}}$ for the adjoint subcategory and $U_{\mathcal{C}}$ for the universal grading group, whose trivial component is $\mathcal{C}_{\mathrm{ad}}$; it is abelian if $\mathcal{C}$ is braided [DGNO, Proposition 2.3, Remark 2.7].

2.1. **Transfer of property F.** Property F passes to direct summands and fusion subcategories. Indeed, if V is a summand of X, naturality of the braiding makes $V^{\otimes n}$ invariant under the braid action on $X^{\otimes n}$, and restriction gives a quotient of $G_{X,n}$. It also passes along a dominant braided tensor functor: the braid image on

the image of an object is a quotient of its original braid image, and every object of the target is a summand of such an image.

If $\mathcal{C}$ has property F, so does $\mathcal{C} \boxtimes \mathcal{C}^{\mathrm{rev}}$. Reversing the braiding inverts the braid generators. On $X \boxtimes Y$, the braid image is a quotient of a subgroup of $G_{X,n} \times G_{Y,n}$, and every object of the product category is a summand of some $X \boxtimes Y$.

2.2. **Pivotal dimensions.** Let $\mathcal{C}$ be pivotal. We write Tr for the right categorical trace, $d_X = \mathrm{Tr}(\mathrm{id}_X)$, and tr for the ordinary matrix trace. For $T \in \mathrm{End}(Z)$,

$$\mathrm{Tr}(T) = \sum_W d_W \,\mathrm{tr}\big(T|_{\mathrm{Hom}(W,Z)}\big).$$

For $T \in \mathrm{End}(Y \otimes Z)$, the right partial trace $\mathrm{ptr}_Z(T) \in \mathrm{End}(Y)$ satisfies $\mathrm{Tr} \circ \mathrm{ptr}_Z = \mathrm{Tr}$, $\mathrm{ptr}_Z((R \otimes \mathrm{id}_Z)T) = R\,\mathrm{ptr}_Z(T)$ and $\mathrm{ptr}_Z(\mathrm{id}_{Y_1} \otimes T_1) = \mathrm{id}_{Y_1} \otimes \mathrm{ptr}_Z(T_1)$. If Y is simple, then $\mathrm{ptr}_Z(T) = (\mathrm{Tr}(T)/d_Y)\mathrm{id}_Y$; here $d_Y \neq 0$, since the dimension of a simple object in a pivotal fusion category is nonzero [EGNO, Proposition 4.8.4].

The function $X \mapsto d_X$ is a character of the fusion ring, so Perron–Frobenius theory gives

$$|d_X| \leq \mathrm{FPdim}(X).$$

For simple X, the left dimension of X is d_{X^*} [La, §2], and $d_{X^*} = \overline{d_X}$ [ENO, Proposition 2.9]. Hence the squared norm of a simple object X is $|X|^2 = d_X d_{X^*} = |d_X|^2$, and $\dim(\mathcal{C}) = \sum_X |d_X|^2 \leq \mathrm{FPdim}(\mathcal{C})$ [ENO, Proposition 8.21], the sum running over the simple objects. For simple X, the scalars $\mathrm{Sl}_X = (d_{X^*}/d_X)\,\mathrm{id}_X$ extend to a tensor automorphism Sl of the identity functor [La, proof of Proposition 2.13], the slope. So if X is simple and $\kappa = d_X/|d_X|$, then $\overline{d_Y}/d_Y = \kappa^{-2n}$ for every simple summand Y of $X^{\otimes n}$, that is,

$$d_Y \in \kappa^n \mathbb{R}. \tag{1}$$

Pivotal structures form a torsor over $\mathrm{Aut}_\otimes(\mathrm{id}_\mathcal{C}) \cong \mathrm{Hom}(U_\mathcal{C}, \mathbb{C}^\times)$ [EGNO, Exercise 4.7.16], [DGNO, Proposition 2.3], and twisting by ψ multiplies d_X by $\psi(\deg X)$. On $\mathcal{C}_{\mathrm{ad}}$ the slope is trivial, so d_Y is real there and does not depend on the pivotal structure. A pseudo-unitary category has a unique pivotal structure with $d_X = \mathrm{FPdim}(X)$, and it is spherical [ENO, Proposition 8.23]. A weakly integral category is pseudo-unitary [ENO, Proposition 8.24].

2.3. **Galois conjugation.** A braided fusion category, possibly with a pivotal structure, has a model over $\overline{\mathbb{Q}}$. The fusion category has a model over a number field [ENO, Theorem 2.30 and Remark 2.33], and braidings and pivotal structures are finitely many solutions of polynomial equations. For $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the conjugate $\mathcal{C}^\sigma$ has the same fusion rules and Frobenius–Perron dimensions. Its braid matrices are obtained by applying σ entrywise, so property F is invariant under Galois conjugation, and centralisers, $\mathcal{C}'$ and $\mathcal{C}_{\mathrm{ad}}$ correspond. If $\mathcal{C}$ is pivotal, so is $\mathcal{C}^\sigma$, with dimensions $\sigma(d_X)$. The squared norm is defined in any fusion category [ENO, §2] and is computed in the $\overline{\mathbb{Q}}$-model, so $\dim(\mathcal{C}^\sigma) = \sigma(\dim \mathcal{C})$. We do not use the form $|d_X|^2$ here, since complex conjugation need not commute with σ.

2.4. **The Frobenius–Perron object.** Let $\mathcal{M}$ be a pivotal braided fusion category and put $S_{XY} = \operatorname{Tr}(c_{Y,X}c_{X,Y})$. The matrix S is symmetric, and for simple X the function $\chi_X(Y) = S_{XY}/d_X$ is a character of the fusion ring [La, §2.2, Proposition 2.4]. For a fusion subcategory $\mathcal{D}$ and a simple object V,

$$V \in \mathcal{D}' \iff S_{YV} = d_Y d_V \quad \text{for every } Y \in \mathcal{O}(\mathcal{D}). \tag{2}$$

This is [DGNO, Proposition C.3], whose normalised matrix $\tilde{s}_{YV}$ equals $S_{YV}/(d_Y d_V)$ by [La, §2.2]; no sphericality is needed.

Proposition 2.1. *Let $\mathcal{M}$ be non-degenerate and pivotal. There is a unique simple object P with $\chi_P = \mathrm{FPdim}$, and*

$$S_{XP} = d_P \,\mathrm{FPdim}(X) \quad \textit{for every simple } X, \tag{3}$$

$$\dim(\mathcal{M}) = |d_P|^2 \,\mathrm{FPdim}(\mathcal{M}). \tag{4}$$

Proof. Non-degeneracy makes the χ_X distinct, so they exhaust the characters [La, §2.4], and FPdim is one of them [EGNO, Proposition 3.3.6]. This gives P and (3). By [La, Corollary 2.20], the matrix S becomes unitary after division by $\sqrt{\dim \mathcal{M}}$ and by a scalar, which has absolute value one by [La, Proposition 2.13]. Hence $SS^\dagger = \dim(\mathcal{M})\, I$, and the row of P gives

$$\dim(\mathcal{M}) = \sum_X |S_{PX}|^2 = |d_P|^2 \sum_X \mathrm{FPdim}(X)^2 = |d_P|^2 \,\mathrm{FPdim}(\mathcal{M}). \qquad \square$$

We call P the *Frobenius–Perron object* of $\mathcal{M}$.

3. The trace argument

Fix an object X of a pivotal braided fusion category for which every $G_{X,n}$ is finite, and let $A_n \subseteq \mathrm{End}(X^{\otimes n})$ be the linear span of $G_{X,n}$. Averaging makes the group action unitary on each multiplicity space. The resulting adjoint makes A_n a finite-dimensional $*$-algebra, with

$$\Big(\sum_{h \in G_{X,n}} \alpha_h h\Big)^* = \sum_{h \in G_{X,n}} \overline{\alpha_h}\, h^{-1}.$$

Thus the involution is independent of the averaged inner products. Tensoring on either side with identity strands preserves it, since it carries braid images and their inverses to braid images and inverses.

The next lemma obtains a trace bound by placing overlapping copies of a unitary on increasingly many strands.

Lemma 3.1 (Trace bound)**.** *Let X be simple in a pivotal braided fusion category, with $G_{X,n}$ finite for every n. Put $d = d_X$ and $f = \mathrm{FPdim}(X)$. Every unitary $u \in A_m$, $m \geq 2$, satisfies*

$$|\operatorname{Tr}(u)| \leq |d| f^{m-1}.$$

Proof. Place k copies $u_1, \dots, u_k$ on consecutive blocks of m strands, successive blocks sharing one strand, so $\ell = k(m-1)+1$. These insertions preserve the involution, so $U = u_1 \cdots u_k$ is unitary. On each multiplicity space its ordinary trace has absolute value at most the dimension of that space. Consequently,

$$|\operatorname{Tr}(U)| = \Big|\sum_W d_W \operatorname{tr}(U|_{\operatorname{Hom}(W, X^{\otimes \ell})})\Big| \le \sum_W N_{X^{\otimes \ell}}^W |d_W| \le \sum_W N_{X^{\otimes \ell}}^W \operatorname{FPdim}(W) = f^\ell.$$

Simplicity of X makes the partial trace of u over its last $m-1$ strands equal to $(\operatorname{Tr}(u)/d)\mathrm{id}_X$. Only u_k meets the last $m-1$ strands of U; tracing them out and iterating gives

$$\operatorname{Tr}(U) = \frac{\operatorname{Tr}(u)^k}{d^{k-1}}, \qquad |\operatorname{Tr}(u)|^k \le |d|^{k-1} f^{k(m-1)+1}.$$

Take k-th roots and let $k \to \infty$. □

Lemma 3.2 (Saturation)**.** *Let X be simple in a pivotal braided fusion category, with $G_{X,n}$ finite for every n. Put $d = d_X$, $f = \operatorname{FPdim}(X)$, $c = c_{X,X}$ and $t = \operatorname{Tr}(c^2)/d$. Then $|t| \le f$, and if $|t| = f$ then $|d| = f$.*

Proof. Since $t = \chi_X(X)$, the bound $|t| \le f$ holds. Assume $|t| = f$ and put $\kappa = d/|d|$. We first bound the total absolute weight of the categorical trace on A_3. Decompose the multiplicity spaces into irreducible representations of the finite group $G_{X,3}$:

$$H_W = \operatorname{Hom}(W, X^{\otimes 3}) \cong \bigoplus_r V_r^{\oplus m_{Wr}}, \qquad A_3 \cong \bigoplus_r \operatorname{End}(V_r),$$

retaining only the constituents that occur. For $T \in A_3$,

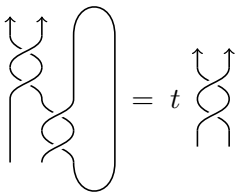
$$\operatorname{Tr}(T) = \sum_r w_r \operatorname{tr}(T|_{V_r}), \qquad w_r = \sum_W m_{Wr} d_W.$$

By (1), $w_r = \kappa^3 v_r$ with v_r real. Let $e \in A_3$ be the central idempotent onto the blocks with $w_r \neq 0$, and let $J \in A_3$ act on V_r by $\operatorname{sgn}(v_r)$ if $w_r \neq 0$ and by 1 otherwise. This is a unitary involution, so Lemma 3.1 gives

$$\sum_r |w_r| \dim V_r = |\operatorname{Tr}(J)| \le |d| f^2. \tag{5}$$

We now exhibit a braid whose trace attains this bound. Let a, b be the images of σ_1, σ_2. Since X is simple, $\operatorname{ptr}_X(c^2) = t\,\mathrm{id}_X$, so tracing over the third strand of a^2b^2 gives $t\,c^2$, and

$$\operatorname{Tr}(a^2b^2) = t \operatorname{Tr}(c^2) = dt^2, \qquad |\operatorname{Tr}(a^2b^2)| = |d| f^2.$$



Partial trace over strand 3.

Since $g = a^2b^2$ is unitary,

$$|d| f^2 = \Big|\sum_r w_r \operatorname{tr}(g|_{V_r})\Big| \le \sum_r |w_r|\,|\operatorname{tr}(g|_{V_r})| \le \sum_r |w_r| \dim V_r \le |d| f^2.$$

Equality holds throughout. Where $w_r \neq 0$, the unitary matrix $g|_{V_r}$ has trace of absolute value $\dim V_r$, and hence is scalar. Hence ge is central in A_3. Since e is central and b^2 commutes with b,

$$0 = [a^2b^2e,\, b] = [a^2e,\, b]\, b^2,$$

so a^2e commutes with b. It also commutes with a, so it is central in A_3. The braid relation gives $b = (ab)\, a\, (ab)^{-1}$, and conjugating the central element a^2e by ab yields $b^2e = a^2e$. Thus $a^2b^{-2}e = e$, so a^2b^{-2} acts trivially on every block of nonzero weight, and

$$\operatorname{Tr}(a^2b^{-2}) = \sum_r w_r \dim V_r = \operatorname{Tr}(\mathrm{id}_{X^{\otimes 3}}) = d^3 = \kappa^3 |d|^3. \tag{6}$$

We compute the same trace directly. Since c^2 has finite order, its inverse has the conjugate trace on each $L_W = \operatorname{Hom}(W, X \otimes X)$, and $d_W = \kappa^4 \overline{d_W}$ by (1). Hence $\operatorname{Tr}(c^{-2}) = \kappa^4 \overline{\operatorname{Tr}(c^2)} = \kappa^4 \bar{t}\bar{d}$. Tracing over the third strand of a^2b^{-2} gives $(\operatorname{Tr}(c^{-2})/d)\, c^2$, so

$$\operatorname{Tr}(a^2b^{-2}) = \frac{\operatorname{Tr}(c^{-2})\operatorname{Tr}(c^2)}{d} = \kappa^4 |t|^2 \bar{d} = \kappa^3 |d| f^2.$$

Comparing with (6) gives $|d| = f$. □

Remark 3.3. The phase κ is needed because pivotal dimensions may be non-real. In a spherical category $\kappa = \pm 1$, and the weights w_r are real. If the chosen pivotal structure has positive dimensions, for example the canonical pivotal structure on a pseudo-unitary category, every weight is nonzero and $e = 1$.

4. Proof of the main result

Proposition 4.1. *Let $\mathcal{M}$ be a non-degenerate pivotal braided fusion category with Frobenius–Perron object P. If $G_{P,n}$ is finite for every n, then $\mathcal{M}$ is pseudo-unitary and P is invertible.*

Proof. By (3), $t = S_{PP}/d_P = \operatorname{FPdim}(P)$, so Lemma 3.2 gives $|d_P| = \operatorname{FPdim}(P) \geq 1$. By (4),

$$\dim(\mathcal{M}) = |d_P|^2 \operatorname{FPdim}(\mathcal{M}) \geq \operatorname{FPdim}(\mathcal{M}) \geq \dim(\mathcal{M}).$$

Hence $\mathcal{M}$ is pseudo-unitary and $\operatorname{FPdim}(P) = |d_P| = 1$. □

Proof of Theorem 1.1. By Proposition 4.1, $\mathcal{C}$ is pseudo-unitary. Let σ be a Galois automorphism. The conjugate $\mathcal{C}^\sigma$ is again non-degenerate and pivotal with property F (§2.3), hence pseudo-unitary. Their common fusion rules give

$$\sigma(\dim \mathcal{C}) = \dim(\mathcal{C}^\sigma) = \operatorname{FPdim}(\mathcal{C}^\sigma) = \operatorname{FPdim}(\mathcal{C}) = \dim(\mathcal{C}).$$

Thus every Galois automorphism fixes $\dim(\mathcal{C})$. As $\dim(\mathcal{C})$ is an algebraic integer [ENO, Theorem 2.3, Remark 2.5], $\operatorname{FPdim}(\mathcal{C}) = \dim(\mathcal{C}) \in \mathbb{Z}$. □

5. Degenerate categories and consequences

To treat a degenerate category $\mathcal{C}$, we place the Frobenius–Perron object of its Drinfeld centre in a subcategory with property F. As in the non-degenerate case, the dimensions of $\mathcal{C}$ need not be real; we need only a sign condition on $\mathcal{C}' \cap \mathcal{C}_{\mathrm{ad}}$, where dimensions are real and independent of the pivotal structure (§2.2).

Let $\mathcal{C}$ be pivotal, and write an object of the centre $\mathcal{Z}(\mathcal{C})$ as (V, γ) with $\gamma_X : V \otimes X \to X \otimes V$. By naturality and monoidality of the pivotal structure a, each a_V is a morphism in $\mathcal{Z}(\mathcal{C})$, so $\mathcal{Z}(\mathcal{C})$ is pivotal and the forgetful functor preserves traces [EGNO, Exercise 7.13.6]. The centre is non-degenerate [DGNO, Corollary 3.9], with $\dim \mathcal{Z}(\mathcal{C}) = \dim(\mathcal{C})^2$ [ENO, Theorem 2.15] and $\mathrm{FPdim}(\mathcal{Z}(\mathcal{C})) = \mathrm{FPdim}(\mathcal{C})^2$ [EGNO, Theorem 7.16.6]. Let

$$\iota_+ : \mathcal{C} \longrightarrow \mathcal{Z}(\mathcal{C}),\ X \longmapsto (X, c_{X,-}), \qquad \iota_- : \mathcal{C}^{\mathrm{rev}} \longrightarrow \mathcal{Z}(\mathcal{C}),\ X \longmapsto (X, c^{-1}_{-,X}),$$

and denote their images by $\mathcal{C}_+$ and $\mathcal{C}_-$. These are fully faithful braided tensor functors [EGNO, Proposition 8.6.1], and ι_+ preserves both dimensions and Frobenius–Perron dimensions [EGNO, Proposition 4.5.7].

Lemma 5.1. *Let $\mathcal{B} = \mathcal{C}_+ \vee \mathcal{C}_-$ and $\mathcal{E} = \iota_+(\mathcal{C}')$. Then $\mathcal{B}' = \mathcal{E}$ and $\mathcal{E}' = \mathcal{B}$. If $\mathcal{C}$ has property F, so does $\mathcal{B}$.*

Proof. The canonical images are mutual centralisers: the double braiding of $\iota_+(X)$ with (V, γ) is $\gamma_X \circ c_{X,V}$, which is the identity for every X precisely when $(V, \gamma) = \iota_-(V)$, and similarly for $\mathcal{C}_-$. Consequently $\mathcal{B}' = \mathcal{C}_+ \cap \mathcal{C}_- = \iota_+(\mathcal{C}')$, since $\iota_+(X) \cong \iota_-(Y)$ forces $X \cong Y$ and $c_{X,Z} = c^{-1}_{Z,X}$ for all Z. The double centraliser theorem gives $\mathcal{E}' = \mathcal{B}'' = \mathcal{B}$ [DGNO, Corollary 3.11]. Finally, the two canonical images centralise each other, so $X \boxtimes Y \mapsto \iota_+(X) \otimes \iota_-(Y)$ defines a braided functor $\mathcal{C} \boxtimes \mathcal{C}^{\mathrm{rev}} \to \mathcal{B}$ [EGNO, Proposition 8.6.1]. It is dominant because these images generate $\mathcal{B}$ and commute up to isomorphism. The transfer properties in §2.1 therefore give property F for $\mathcal{B}$. □

Lemma 5.2. *Let $\mathcal{C}$ be a pivotal braided fusion category with $d_Y > 0$ for every simple $Y \in \mathcal{C}' \cap \mathcal{C}_{\mathrm{ad}}$. Then $\mathcal{C}$ has a pivotal structure with $d_Y = \mathrm{FPdim}(Y)$ for every simple $Y \in \mathcal{C}'$.*

Proof. The symmetric category $\mathcal{C}'$ is integral [DGNO, Corollary 2.46], hence pseudo-unitary, with a pivotal structure of dimensions FPdim [ENO, Propositions 8.23, 8.24]. The restricted pivotal structure differs from it by a character η of $U_{\mathcal{C}'}$, so $d_Y = \eta(\deg Y)\,\mathrm{FPdim}(Y)$ for simple $Y \in \mathcal{C}'$. Inclusion induces a homomorphism $\phi : U_{\mathcal{C}'} \to U_{\mathcal{C}}$, and since the universal grading is faithful with trivial component $\mathcal{C}_{\mathrm{ad}}$, the kernel of ϕ consists of the degrees of the simple objects of $\mathcal{C}' \cap \mathcal{C}_{\mathrm{ad}}$. For $g \in \ker \phi$, choose a simple $Y \in \mathcal{C}'$ of degree g; then $Y \in \mathcal{C}' \cap \mathcal{C}_{\mathrm{ad}}$ and $\eta(g) = d_Y / \mathrm{FPdim}(Y) > 0$. Since $U_{\mathcal{C}'}$ is finite, the values of η are roots of unity, so positivity makes η trivial on $\ker \phi$. Thus η factors through ϕ and extends to a character ψ of the finite abelian group $U_{\mathcal{C}}$. Twisting the pivotal structure by ψ^{-1} gives $d_Y = \mathrm{FPdim}(Y)$ on $\mathcal{C}'$ (§2.2). □

Theorem 5.3. *Let $\mathcal{C}$ be a pivotal braided fusion category with property F such that $d_Y > 0$ for every simple object Y of $\mathcal{C}' \cap \mathcal{C}_{\mathrm{ad}}$. Then $\mathcal{C}$ and all its Galois conjugates are pseudo-unitary, and* $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$.

Proof. By Lemma 5.2 we may assume that $d_Y = \mathrm{FPdim}(Y)$ for simple $Y \in \mathcal{C}'$. Set $\mathcal{M} = \mathcal{Z}(\mathcal{C})$, let $\mathcal{B}$ and $\mathcal{E}$ be as in Lemma 5.1, and let P be the Frobenius–Perron object of $\mathcal{M}$. For every simple $Y \in \mathcal{E}$ we have $d_Y = \mathrm{FPdim}(Y)$, and (3) gives

$$S_{YP} = d_P \,\mathrm{FPdim}(Y) = d_Y d_P.$$

The centraliser criterion (2) gives $P \in \mathcal{E}' = \mathcal{B}$. Since $\mathcal{B}$ has property F, all braid images of P are finite, and Proposition 4.1 shows that $\mathcal{M}$ is pseudo-unitary. Thus $\dim(\mathcal{C})^2 = \mathrm{FPdim}(\mathcal{C})^2$, and as both are positive, $\mathcal{C}$ is pseudo-unitary.

Let σ be a Galois automorphism. The conjugate $\mathcal{C}^\sigma$ is pivotal with property F, and its dimensions on $(\mathcal{C}^\sigma)' = (\mathcal{C}')^\sigma$ are $\sigma(\mathrm{FPdim}(Y)) = \mathrm{FPdim}(Y)$, since $\mathcal{C}'$ is integral. So $\mathcal{C}^\sigma$ satisfies the hypothesis and is pseudo-unitary. As in the proof of Theorem 1.1, $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$. □

The hypothesis holds when $\mathcal{C}$ is non-degenerate, and more generally when $\mathcal{C}' \cap \mathcal{C}_{\mathrm{ad}} = \mathrm{Vec}$. It also holds, after conjugation, when $\mathcal{C}$ has a pseudo-unitary Galois conjugate, and then no pivotal structure need be given.

Corollary 5.4. *Let $\mathcal{C}$ be a braided fusion category with a pseudo-unitary Galois conjugate. If $\mathcal{C}$ has property F, then $\mathcal{C}$ is pseudo-unitary and weakly integral.*

Proof. A pseudo-unitary conjugate $\mathcal{C}^\sigma$ with its canonical spherical structure has $d_Y = \mathrm{FPdim}(Y) > 0$ for every simple Y, and it has property F. By Theorem 5.3 it is weakly integral, hence so is $\mathcal{C}$, which is then pseudo-unitary [ENO, Proposition 8.24]. □

The theorem also has an objectwise consequence. To apply it to the fusion subcategory $\langle X \rangle$ generated by an object X, we must establish property F for every object of that subcategory. The following lemma shows that finiteness of all $G_{X,n}$ suffices.

Lemma 5.5 (Closure)**.** *Let X be an object of a braided fusion category. If $G_{X,n}$ is finite for every n, then $\langle X \rangle$ has property F.*

Proof. The case $X = 0$ is immediate. Otherwise, summands of finite sums of positive powers of X form a nonzero full abelian subcategory closed under tensor products and subquotients. By [EGNO, Corollary 4.11.4], it contains the unit and is rigid, so it equals $\langle X \rangle$. Thus every object of $\langle X \rangle$ is a summand of some $Z = \bigoplus_i X^{\otimes k_i}$, with $k_i \geq 1$. For each $n \geq 2$, the pure braid group P_n preserves each summand $X^{\otimes k_{i_1}} \otimes \cdots \otimes X^{\otimes k_{i_n}}$ of $Z^{\otimes n}$. By cabling, its action there has image in the finite group $G_{X,k_{i_1}+\cdots+k_{i_n}}$. Thus its image on $Z^{\otimes n}$ is finite, being contained in a finite product of finite groups. Since $[B_n : P_n] = n!$, the full braid image is finite as well [ERW, Lemma 2.1]. Naturality passes finiteness to every summand of Z, as in §2.1. □

Consequently, the dimension of a particular simple object can force its own braid images to be infinite.

Corollary 5.6. *Let X be a simple object of a braided fusion category, and suppose that $\langle X \rangle$ is non-degenerate and pivotal, or has a pseudo-unitary Galois conjugate. If $\mathrm{FPdim}(X)^2 \notin \mathbb{Z}$, then $G_{X,n}$ is infinite for every sufficiently large n.*

Proof. If every $G_{X,n}$ were finite, Lemma 5.5 together with Theorem 1.1 or Corollary 5.4 would make $\langle X \rangle$ weakly integral. Then $\mathrm{FPdim}(X)^2 \in \mathbb{Z}$ by [ENO, Proposition 8.27], a contradiction. Hence G_{X,n_0} is infinite for some n_0. Tensoring with a nonzero object is faithful, so tensoring with identity strands embeds G_{X,n_0} in $G_{X,n}$ for every $n \geq n_0$. □

In particular, the conclusion holds in every unitary braided fusion category, since its fusion subcategories are unitary and hence pseudo-unitary. The simplicity hypothesis is essential: direct sums in a weakly integral category need not have integral squared Frobenius–Perron dimension.

Acknowledgements

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